

Arabizi-Arabic Neural Transliteration System

Bridging the Gap Between Romanized &
Native Arabic Text

Problem



The Arabizi Problem

- Arabic script is not compatible with standard keyboards
- Social media users use Roman letters (Arabizi)
- Romanization depends on local dialect and colonial language

Intelligence & Security Implications

- OSINTers struggle to search Arabic social media
- Terrorist/hate speech monitoring misses Arabizi content
- LE cannot query Arabic databases with Romanized input

Current Tools Gap

- Existing transliteration tools are rule-based and inaccurate
- No dialect aware bidirectional neural approaches conversion tools
- Need for real-time, accurate transliteration at scale

Solution

Arabizi-Arabic Transliterator

- A Transformer-based neural transliteration system
- Bidirectional conversion: Arabizi ↔ Arabic
- Real-time API with web interface

Key Features

- Sequence-to-sequence architecture (Encoder-Decoder)
- Beam search decoding for high-quality outputs
- Handles diacritics, special characters, and Unicode normalization
- Returns top-k predictions for confidence scoring

The screenshot shows a web interface for the 'Arabizi -- Arabic Translator'. At the top, there's a title bar with a logo and the text 'Arabizi -- Arabic Translator'. Below this, there are two dropdown menus: the first is set to 'Arabic → Arabizi' and the second is set to 'Palestinian Arabic'. The main area is split into two columns. The left column, labeled 'Arabic', contains a text input field with the Arabic word 'سلام' (Salaam) and a 'Type Arabic text' label. The right column, labeled 'Arabizi', displays a list of six transliteration suggestions: '1. salaam', '2. salam', '3. saalim', '4. silmaam', '5. s2aalam', and '6. sal2aam'. Above this list is a small text indicating 'All returned transliterations · 0:08'. At the bottom of the interface, there are three buttons: 'Clear', 'Copy Results', and 'Transliterate'. Below these buttons is a section titled 'Dialect options' which lists various Arabic dialects: 'Palestinian Arabic' (marked as 'Available'), 'Levantine Arabic' (marked 'Coming later'), 'Egyptian Arabic' (marked 'Coming later'), 'Gulf Arabic' (marked 'Coming later'), 'Iraqi Arabic' (marked 'Coming later'), and 'Maghrebi Arabic' (marked 'Coming later').

High-Level Architecture

Architecture Components

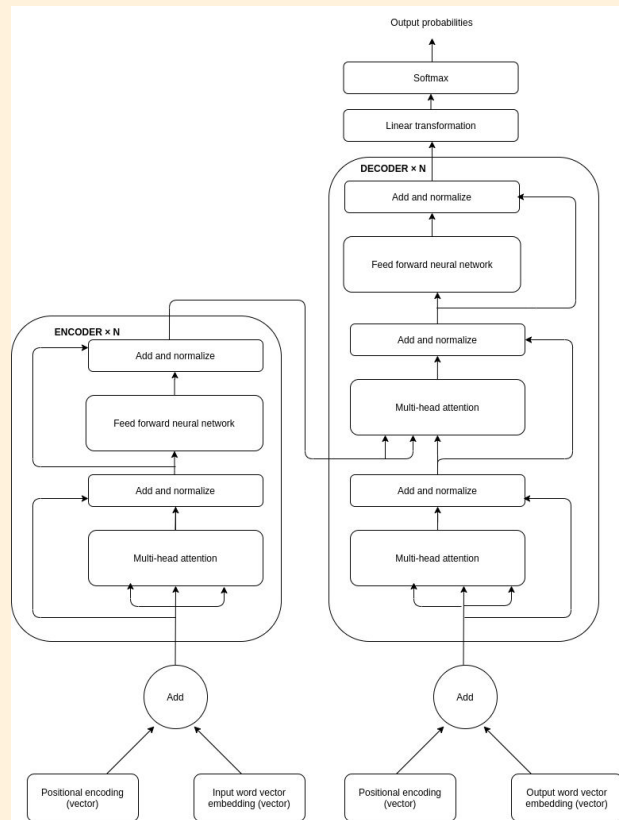
- Transformers model
- Encoder: 5 layers, 8 attention heads
- Decoder: 5 layers, 8 attention heads
- Embedding dimension: 320
- Feed-forward dimension: 1280
- Dropout: 20%

Input Processing

- Character-level tokenization
- Special tokens: <PAD>, <BOS>, <EOS>, <UNK>
- Unicode normalization (NFC)
- Lowercase normalization for Arabizi

Output Generation

- Beam search (width=20) with length penalty
- Returns top-10 predictions per input



Architecture Deep Dive

Vocabulary

- Combined character vocabulary for both scripts
- ~150 tokens covering Arabic alphabet + Roman letters + digits + punctuation

Training Configuration

- Loss: Cross-entropy with label smoothing
- Optimizer: Adam with custom learning rate schedule
- Batch size: Optimized for GPU memory
- Mixed precision training for efficiency

Inference Pipeline

- FastAPI backend for low-latency serving
- CUDA GPU acceleration when available
- Fallback to CPU for portability

Model Bundle:

- Includes tokenizer, model weights, and config
- Self-contained .pt bundle files

Result Re-Ranker

Why Re-Ranking

- Arabizi is highly ambiguous: one Romanized input can map to multiple Arabic words
- Beam search returns strong candidates, but the top is not always the best user-facing answer
- Re-ranking lets the system use extra signals beyond raw sequence likelihood

Re-Ranking Model:

- Lightweight MLP re-ranker trained to score beam-search candidates
- Runs after the Transformer generates top candidate transliterations
- Produces a cleaner top-k list for the web demo and API

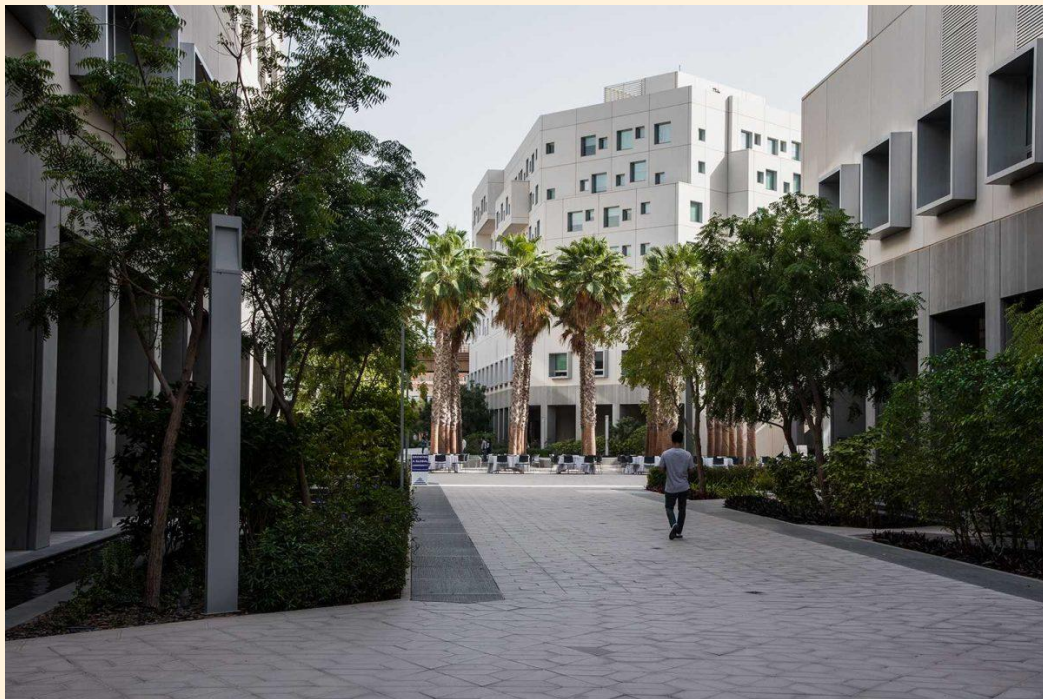
Candidate Features:

- Beam rank and score gap from the best candidate
- Source and candidate length ratios
- Digit usage for Arabizi letters like 3, 7, 2, 5
- Space, vowel-like, and script-character patterns

Impact:

- More useful alternatives for human review
- Keeps inference efficient because the re-ranker is small

Training and Data Preprocessing



Dataset: Maknuune v1.0.1

- From NYU Abu Dhabi
- Parallel Arabizi-Arabic sentence pairs
- Diverse sources: social media, news, forums
- 36k observations

Preprocessing Steps:

- Unicode normalization (NFC form)
- Arabic tatweel (kashida) removal
- Whitespace normalization
- Text stripping and cleaning
- Split: Training / Validation / Test sets

Model Performance

Metrics

Character Error Rate (CER)

Exact match after normalization

Direction-specific evaluation: Arabizi → Arabic and Arabic → Arabizi

Evaluation Results

Greedy decoding on 300 random cleaned Maknuune entries

- Arabizi → Arabic: 91.3% normalized exact match, 0.032 avg normalized CER
- Arabic → Arabizi: 92.7% normalized exact match, 0.026 avg normalized CER

Beam Search & Re-Ranking

- Arabizi → Arabic: 93.3% top-1 normalized exact match
- Arabizi → Arabic: 98.3% gold answer appears in top 3
- Arabic → Arabizi: 91.7% top-1 normalized exact match

Robustness

- Handles common Arabizi phonemes such as 3, 7, and 2
- Returns alternatives when an input is ambiguous
- Graceful degradation with unknown characters and non-standard spellings



Arabizi → Arabic

Palestinian Arabic

Arabizi

Type Arabizi text

salaam

Arabic

All returned transliterations · 0:08

1. سلام
2. صلام
3. لسام
4. ئلام
5. سلامة
6. سلامي

Clear

Copy Results

Transliterate

Dialect options

Palestinian
Arabic

Available

Levantine
Arabic

Coming
later

Egyptian
Arabic

Coming
later

Gulf Arabic

Coming later

Iraqi Arabic

Coming later

Maghrebi
Arabic

Coming
later



Arabic → Arabizi



Palestinian Arabic



Arabic

Type Arabic text

سلام

Clear

Copy Results

Transliterate

Dialect options

Palestinian
Arabic

Available

Levantine
Arabic

Coming
later

Egyptian
Arabic

Coming
later

Gulf Arabic

Coming later

Iraqi Arabic

Coming later

Maghrebi
Arabic

Coming
later

Arabizi

All returned transliterations · 0:07

1. salaam
2. salam
3. saalim
4. silmaam
5. s2aalam
6. sal2aam



Arabizi → Arabic

Palestinian Arabic

Arabizi

Type Arabizi text

keef

Arabic

All returned transliterations · 0:07

1. كيف
2. كيفة
3. كافة

Clear

Copy Results

Transliterate

Dialect options

Palestinian
Arabic

Available

Levantine
Arabic

Coming
later

Egyptian
Arabic

Coming
later

Gulf Arabic

Coming later

Iraqi Arabic

Coming later

Maghrebi
Arabic

Coming
later



Arabic → Arabizi



Palestinian Arabic



Arabic

Type Arabic text

كيف

Clear

Copy Results

Transliterate

Dialect options

Palestinian
Arabic

Available

Levantine
Arabic

Coming
later

Egyptian
Arabic

Coming
later

Gulf Arabic

Coming later

Iraqi Arabic

Coming later

Maghrebi
Arabic

Coming
later

Arabizi

All returned transliterations · 0:07

1. kafi
2. kiif
3. fiik
4. keef
5. ykaf
6. kafa



Arabizi → Arabic



Palestinian Arabic



Arabizi

Type Arabizi text

qahwe

Arabic

All returned transliterations · 0:09

1. قهوة
2. قحوة
3. حقوة
4. قهوةٲة
5. قهوةزة

Clear

Copy Results

Transliterate

Dialect options

Palestinian
Arabic

Available

Levantine
Arabic

Coming
later

Egyptian
Arabic

Coming
later

Gulf Arabic

Coming later

Iraqi Arabic

Coming later

Maghrebi
Arabic

Coming
later



Arabic → Arabizi



Palestinian Arabic



Arabic

Type Arabic text

قهوة

Clear

Copy Results

Transliterate

Dialect options

Palestinian
Arabic

Available

Levantine
Arabic

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later

Egyptian
Arabic

Coming
later

Gulf Arabic

Coming later

Iraqi Arabic

Coming later

Maghrebi
Arabic

Coming
later

Arabizi

All returned transliterations · 0:07

1. qahwe
2. qahwwe
3. hwuqkshe
4. waqahwe
5. hwuqkkshe
6. qahwekahwe

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https://github.com/kagoeckner/arabizi_arabic_transliterator

https://kagoeckner.github.io/arabizi_arabic_transliterator/